

Q1:

A system takes raw attendance data of students, processes it, and generates a percentage report. Explain how this scenario represents the working of a computer system.

Answer:

This scenario reflects the basic working cycle of a computer system: input, processing, and output. The raw attendance data acts as input, which is processed using calculations to determine attendance percentage. The final report is the output, representing meaningful information. This demonstrates how a computer converts raw data into useful information using digital processing.

Q2:

Different users require different types of computing devices. Analyze the requirements of a student, a graphic designer, and a traveler, and identify the most suitable type of computer for each.

Answer:

A student typically requires a desktop or laptop for general tasks like assignments and browsing. A graphic designer needs a high-performance workstation due to demanding applications such as image rendering and design software. A traveler prefers a lightweight notebook or tablet for portability and convenience. This shows that computers are designed based on performance needs and mobility requirements.

Q3:

Modern computers are smaller, faster, and more efficient compared to early computers. Explain how technological advancements across different generations made this transformation possible.

Answer:

Initially, computers used vacuum tubes, which were large and inefficient. The introduction of transistors reduced size and increased reliability. Integrated circuits further miniaturized components, while microprocessors combined multiple functions into a single chip. These advancements improved speed, reduced power consumption, and made computers

compact. The evolution across generations reflects continuous improvement in hardware technology.

Q4:

While working on a computer, a user types text, clicks icons, and scans documents. Explain how different input devices are involved in this process and their specific roles.

Answer:

Typing text involves the keyboard, which inputs characters into the system. Clicking icons is done using a mouse, which controls the pointer and allows interaction with graphical elements. Scanning documents uses a scanner to convert physical documents into digital form. Each device serves a specific purpose, enabling different types of data input into the computer system.

Q5:

Computers operate using only two digits (0 and 1), yet they can represent complex data like images, text, and videos. Explain how this is possible.

Answer:

Computers use the binary number system, where data is represented using bits (0 and 1). Groups of bits form bytes, which represent characters through encoding systems like ASCII or Unicode. More complex data such as images and videos are stored as combinations of binary patterns. This structured representation allows computers to process and store all types of data despite using only two digits.

Q6:

A company initially stores all employee data in a single table but later faces issues in managing and updating records. Explain why this approach is inefficient and how database design can be improved.

Answer:

Storing all data in a single table leads to redundancy, inconsistency, and difficulty in managing large datasets. This is known as a flat-file structure. A better approach is to use a relational database, where data is divided into multiple related tables. Relationships between tables reduce duplication and improve data integrity. This makes the system more efficient and easier to maintain.

Q7:

A user sends emails, watches videos online, and downloads files. Explain how these activities are supported by different Internet services.

Answer:

Sending emails uses electronic mail services for communication. Watching videos involves web services and streaming technology through the World Wide Web. Downloading files uses protocols such as FTP. These services work together to provide different functionalities over the Internet, enabling communication, entertainment, and data transfer.

Q8:

When a user enters a URL in a browser, a webpage is displayed within seconds. Explain the process that takes place between the browser and the server.

Answer:

When a URL is entered, the browser sends a request to a web server using HTTP. The server processes the request and sends back the webpage data, usually in HTML format. The browser then interprets this HTML and displays it visually. This interaction between browser and server allows users to access information from anywhere in the world.

Q9:

A computer slows down significantly when multiple applications are running. Explain this behavior in terms of CPU and memory usage.

Answer:

The CPU is responsible for processing instructions, while RAM stores active programs and data. When multiple applications are running, both CPU and memory resources are heavily used. If memory becomes insufficient, the system slows down because it cannot efficiently handle all tasks. This results in reduced performance and slower response time.

Q10:

An organization needs to search records, sort data, and generate reports regularly. Explain how a Database Management System (DBMS) supports these operations.

Answer:

A DBMS provides tools for storing, organizing, and retrieving data efficiently. It allows users to query data to find specific records, sort data based on different criteria, and generate reports for analysis. These features make it easier to manage large amounts of data and support decision-making processes within the organization.